DESIGN THINKING · EMPATHISE & DEFINE

The Research Behind Plannr

Stakeholder study and problem definition for the college timetable system

1. Project Topic

This topic was not one of the areas discussed in class. It came out of my own frustration as a student. Working out what was on, where it was and whether it had moved always meant checking several places at once: the timetable website for classes, my inbox for cancelled lectures and changed classrooms, and separate notices for test dates and assignment deadlines. Nothing showed a single day in full.

The question the project set out to answer was therefore:

> **How do we reduce all the manual work a student has to do to know what class, test or assignment is on, when it is, where it is and in what mode?**

The study below is the research that **Plannr** was built from. The working system is live at plannr.paraswtf.com, and signing in needs nothing more than a college Google account.

Issues identified in the current timetable website:

- Too difficult to use on a mobile phone.
- Loading times are long.
- There is no installable app.
- It does not work without an internet connection.
- It is optimised only for clicking, with no touch, swipe or keyboard interaction.
- There are no personalised schedules; students must search the whole college timetable to find their own group's classes.
- Class conflicts are not rejected when coordinators add clashing slots.
- A coordinator can edit an event, but the system tells nobody, so the same change has to be sent again as a mail.
- Cancelled lectures, changed classrooms and test dates are announced by email, so students have to watch an inbox as well as the timetable.
- Classes, tests and assignments live in separate places, so no single view answers what is on, where it is and in what mode.
- Rooms can be booked without checking what the session needs, so a class can land in a room with no projector or whiteboard.

2. Stakeholders

A stakeholder is anyone who affects, or is affected by, the timetable. The list was built by following one timetable through its life: who sets it up, who makes it, who uses it and who plans around it.

#	Stakeholder	Interest / stake in the timetable
1	Students	Check the timetable and their mail each day to know where to be. Attendance depends on acting on it correctly.
2	Faculty / subject teachers	Teach several groups, which may be whole divisions or students from different divisions on the same major or minor. They need rooms that have what the session requires.
3	Class / timetable coordinators	Create and revise the timetable and edit individual events. Today they also have to mail out the changes they have already made.
4	Administration (admins)	Set up student accounts, groups, courses and batches, and record once what each room has.

3. Empathy Maps

Techniques used: students were **observed** opening the timetable on their own phones on campus; short **interviews** were held with 4 students, 2 faculty members, 1 coordinator and 1 administration staff member; a small **survey** was circulated among students in my year; and a brief **focus group** with a mixed set of users was held so that each group heard the others' constraints. Throughout, questions were kept open ("show me what you did last time") and no solution was suggested, so that what was recorded was what users actually feel and do.

Stakeholder	Says	Thinks	Does	Feels
Student	"I checked the timetable, but the cancellation was in a mail I opened later."	There should be one place for all of it.	Checks the timetable and mail each day, and still misses a class or a cancellation now and then.	Annoyed at turning up for a class that was called off.
Faculty member	"I teach several groups, not just one division." "The room had no projector."	The room should already have what my session needs.	Notes the day's classes down at the start of the day from the main page and mail, and mails the group when a class is cancelled.	Irritated at doing the same lookup and the same mail every time.
Coordinator	"I change the event, then I still have to mail everyone about it."	If I have already edited it here, the mail should send itself. And it should warn me before I save a clash.	Edits the event in the timetable, then writes the same change out again as a mail.	Doing the same work twice, and never sure everyone saw the mail.
Administration	"They booked a room with no projector and then asked to move."	What each room has should be recorded once and used by the system.	Sets up accounts, groups, courses and batches, and keeps a separate note of what each room has.	Tired of fixing bookings that should not have been allowed.

4. Defining the Problem Statement

4.1 Space Saturation and Group

Observations and quotes were written on notes and grouped by affinity on a wall until four clusters settled.

	Cluster	Observations grouped under it	Insight
C1	Access on the phone	Pinch-zoom · sideways scrolling · click-only · no touch or keyboard support · almost everyone uses a phone	The site was designed for a device nobody actually uses it on.
C2	Speed and availability	Long waits on mobile data · whole college table loaded every visit · fails in labs and basements · no installable app	It is least available exactly where and when it is most needed.
C3	Relevance and scattered information	Searching the whole college timetable for one group · cancellations and room changes sent by mail · test dates announced separately	No single view answers what is on, where and in what mode, so the student assembles it by hand.
C4	Correctness and process	Clash found by a phone call · no validation on save · rooms booked without checking what the session needs · event edited, then mailed out again by hand	The system records a change but does not act on it, so people finish the job themselves.

4.2 The Four Ws

WHO	Students, faculty and timetable coordinators mainly; also the admins who set up the accounts, groups, courses and rooms it all runs on.

WHAT	They cannot get one view of what is on, where and in what mode; the answer is assembled by hand from the timetable, mail and separate notices. Coordinators can save clashing classes, rooms can be booked without the equipment a session needs, and changes do not reach people automatically.
WHERE	On mobile phones, on campus, including labs and other weak-network areas, mostly at the start of a semester and after every revision.
WHY	Because acting on stale information means missed classes, classes attended after they were called off, wasted preparation and sessions held in rooms that lack what they need.

4.3 The Five Whys

1	**Why** do students turn up for classes that were cancelled? Because the cancellation reached them only as a mail, and it was not seen in time.
2	**Why** was it only a mail? Because the coordinator edits the event in the system, but the system tells nobody, so they send a mail as well.
3	**Why** does the system tell nobody? Because it publishes one page for everybody and has no idea which students a change affects.
4	**Why** does it not know? Because the timetable is stored as a published document, not as data linked to students and their groups.
5	**Why** was it built that way? Because it was treated as a notice board, not a system that answers and checks.
	Root cause: the timetable is modelled as a document to be published instead of validated, personalised data to be delivered. That is why a change entered in one place still has to be announced by hand, and why every issue in Section 1 follows from this one decision.

4.4 Problem Statement

How do we reduce all the manual work a student has to do to know what class, test or assignment is on, when it is, where it is and in what mode?

Today that answer is put together by hand from the timetable, mail and separate notices for tests and assignments. **Students and faculty need one place that shows their own day in full, and the system needs to do the rest itself: refuse clashing classes, offer only rooms that meet what the session needs, and announce a change to everyone it affects, so that a coordinator acts in one place and nothing more is needed.**

Outcome: the system built from this problem statement is live at plannr.paraswtf.com, open with a college Google account.